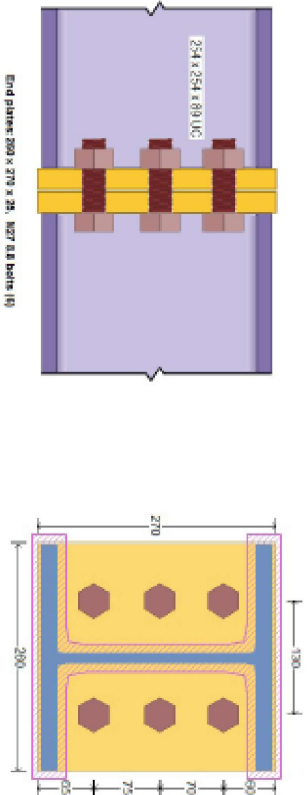


End plate splice calculation

Beam: Lower Ground Spine Beam
Beam span: 6.6 m. Section: 254 x 254 x 89 UC S355



Splice 1: 2.2 m. from R1. BM: 126.3 kNm SF: 29.10 kN (factored)

Section size: 254 x 254 x 89 UC S355

Section dims: D = 260.3 B = 256.3 T = 17.3 t = 10.3 d = 200.3 r = 12.7

Use S275 end plates 260 x 270 x 25 mm

Use 3 pairs of M27 8.8 bolts at 130 mm cross centres; top of plate to top row of bolts: 60; second row: 130 mm; rows 3- spaced at 75 mm

Min fillet weld sizes:

Top (compression) flange: nominal 8 mm (bearing fit assumed)
Bottom (tension) flange: 17.3/(2 x 0.7) = 12.4 mm, say 13 mm (full strength)
Web: 10.3/(2 x 0.7) = 7.4 mm, say 8 mm (full strength)
N.B. 13mm welds may not be practicable

Basic detailing checks

Using M27 bolts, washer diameter 50 mm

Check plate height (270 mm) >= section depth (260 mm) OK

Check top bolt washer clears top flange weld: top of plate to top bolt (60 mm) >= 56 mm OK

Check bottom bolt washer clears bottom flange weld: top of plate to bottom bolt (205 mm) <= 209 mm OK

Check bolt washers clear web welds: bolt cross centres (130 mm) >= 77 mm OK

Check bolt cross centres >= 2.5D (67.5 mm) OK

Check bolt vertical centres (70 mm) >= 2.5D (67.5 mm) OK

Bolt edge distance (65.0 mm) >= 1.25D (33.8 mm) OK

Top bolt end distance (60 mm) >= 1.25D (33.8 mm) OK

Bottom bolt end distance (65 mm) >= 1.25D (33.8 mm) OK

All dimensional checks satisfied

End plate beam splice

Section size: 254 x 254 x 89 UC

Basis of design

This calculation is derived from the SCI publication : Design of Moment Connections. The compression zone is taken as acting at the centre of the top flange. The top pair of bolts act in shear only. The bending moment at the splice is resisted by the other bolt rows. The tensile capacity of each bolt pair is the least value of:

- * The capacity of the bolt row acting alone [Table 2.5]
- * Other rows: the capacity of this bolt row and the row(s) below it acting as a group less the capacity of the row(s) below it acting as a group (2.3.1)

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* The tensile capacity of the beam web (row(s) above bottom row)

Bolt bearing capacity on plate = 27 x 25 x 460/1000 = 310.5 kN

Bolt shear capacity = 459 x 375/1000 = 172 kN <<<

Enhanced bolt tension capacity, P_tⁱ = 459 x 560/1000 = 257.0 kN

Max tensile length of web per bolt row, L_w = (130/2) x tan 60 = 113 mm above and below each row

N.B. the length below may not extend into the length above from the row below

Web tensile capacity per mm = 10.3 x 345/1000 = 3.55 kN/mm

m₁, bolt centre to 20% into web weld = 53.5 mm (8mm fillet welds assumed)

m₂, bolt centre to 20% into bottom flange weld = 32.4 mm (13mm fillet welds assumed)

λ₁ = m₁/(m₁ + e) = 0.45 λ₂ = m₂/(m₁ + e) = 0.27 α = 2t (P207 Appendix iii)

n, effective edge distance = min (edge distance, 1.25m) = 65.0 mm

L_{er}(i) = 2αm = 335.8mm L_{er}(ii) = 4m + 1.25e = 295.0mm L_{er}(iii) = αm = 335.8mm (Table 2.4, α = 2t)

L_{er} for bolts acting alone (Table 2.5)

Other rows: L_{er} = min(max(ii+iii)/2, ii, j) = 315.4 mm

Bottom row: L_{er} = min(ii, j) = 295.0 mm

Non-bottom end row: L_{er} = max(ii/2, iii/2) + p/2 = 205.4 mm

L_{er} = ii/2 + p/2 = 185.0 mm

M_p = L_{er} x t² x p_y/4

Mode 1 (end plate yields at bolt positions and either side of web): P_t = 4M_p/m

Mode 2 (end plate yields either side of web; bolts fail in tension): P_t = (2M_p + nCP_t)/m + n)

Mode 3 (bolt tensile failure): P_t = ΣP_tⁱ

Bolt rows are numbered from the bottom (bottom pair = 1; top pair = 3)

Row	L _{er}	M _p	Mode 1	Mode 2	Mode 3	L _{web}	P _{t,web}	P _{t,group}	P _{t,kN}
1	315	13,061	977	503	514				502.6<<<
2	295	12,217	914	488	514	225	800		488.4
[1+2]	390	16,167	1,210	837	1,028				837
[1+2]-1									334.5<<<

Check moment and shear capacity of splice

Compressive resistance of top flange = 1.4 x 345 x 256.3 x 17.3/1000 = 2142 kN

Bolt line	Bolt L.A. mm	P, kN	Row RM kNm	Bolt F _y /P _y	F _y , kN
1	192	502.6	96.3	0.40	137.7
2	116	*305.8	35.6	0.40	137.7
3	46	0.0	0.0	1.00	344.3
Totals		808.4	131.9		619.7

* tension reduced to be within elastic limit

Section ultimate moment capacity = P_yS_x = 421 kNm; Splice capacity = 131.9 kNm (31.3%)

Splice 1: Design moment: 126.3 kNm Design SF: 29.1 kN

No further checking necessary

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URBAN REGEN LONDON Ltd.

TITLE	ADDRESS	DATE
PROPOSED DRAWINGS	95 MALVERN ROAD LONDON, NW6	MARCH 2019
CLIENT	SCALE	REV.
Mr. C. NORMAN	1:50 @ A3	
	DRAWING NO. 190328/33	